ShopSmart Grounded Product Assistant

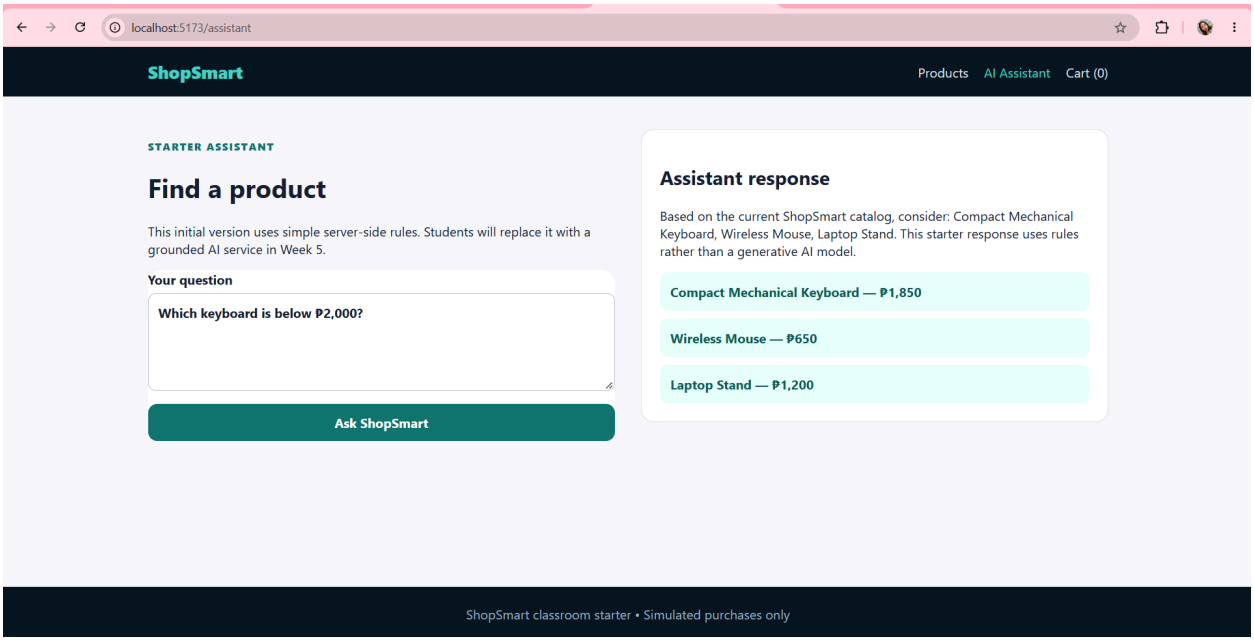
Screenshots and Test Evidence

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PDC50
5 November 2026

Baseline (before changes)

Action	Asked "Which keyboard is below ₱2,000?" on the starter assistant.
Observed	Returned the Compact Mechanical Keyboard, but also the Wireless Mouse and Laptop Stand as filler.
Why it matters	This was my first failing case. The starter had no relevance check.

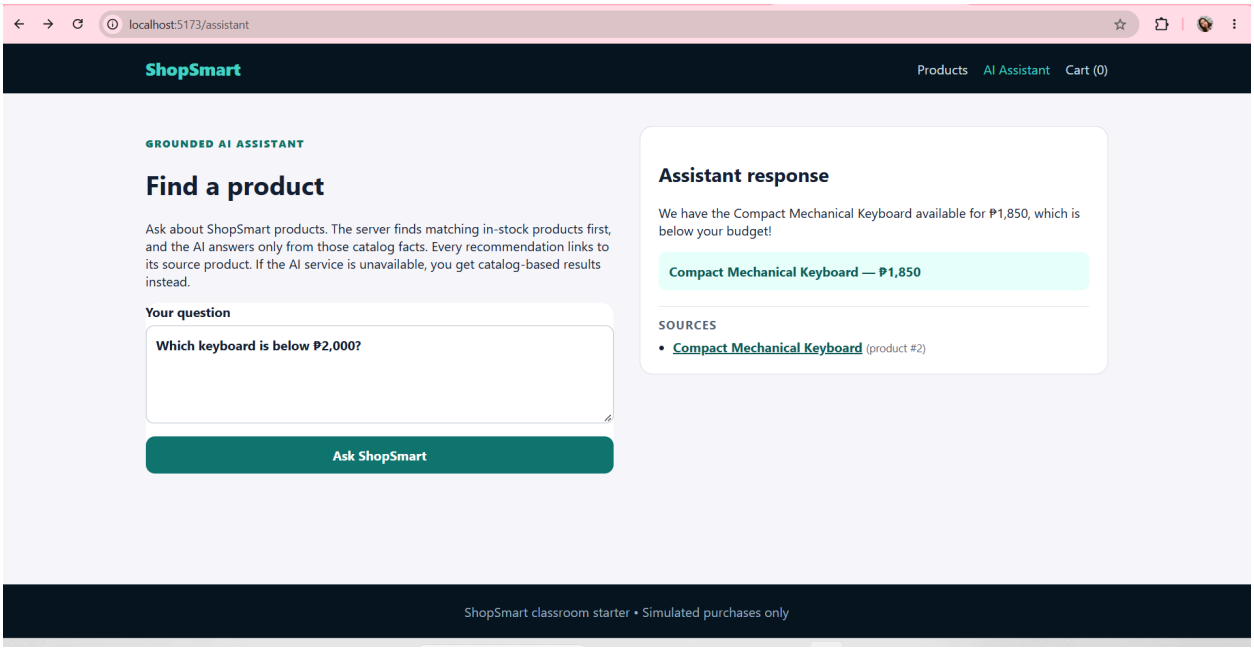
Screenshot:



Case 1: Catalog match

Action	Asked "Which keyboard is below ₱2,000?"
Expected	A real in-stock keyboard from the catalog, with its real price and a working link.
Result	Pass. Only the Compact Mechanical Keyboard (id 2, ₱1,850), with a product link and a Sources list.

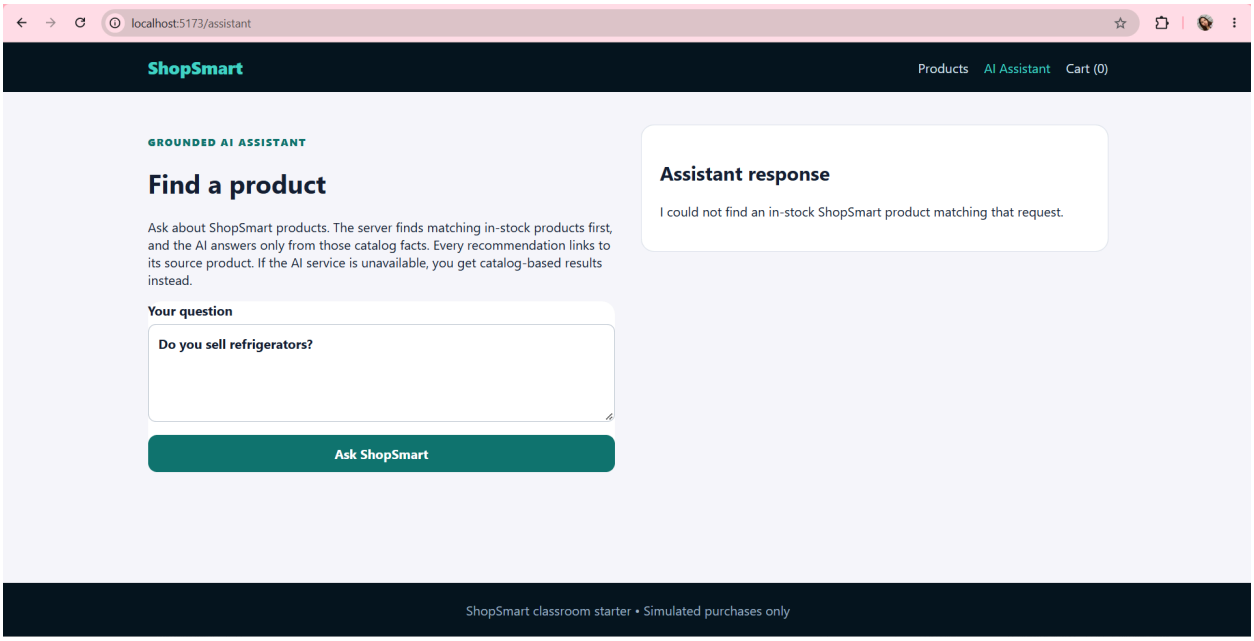
Screenshot: assistant answer with Sources



Case 2: No match

Action	Asked "Do you sell refrigerators?"
Expected	Honest no-match answer. No made-up product or source.
Result	Pass. "I could not find an in-stock ShopSmart product matching that request." No products, no sources, Gemini not called.

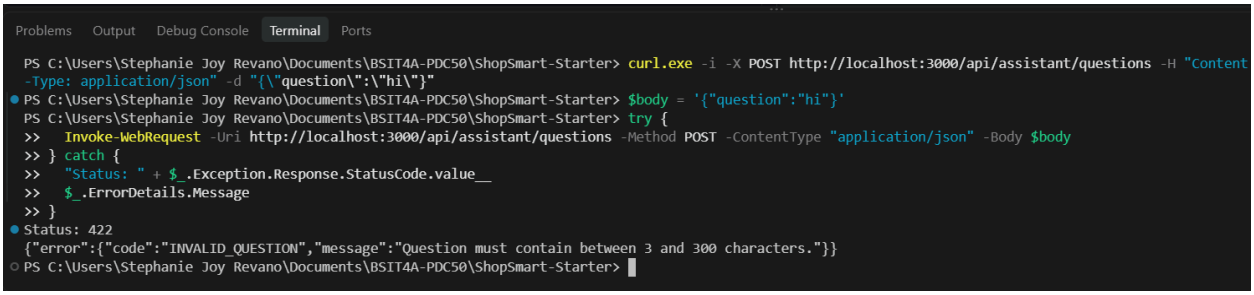
Screenshot: no-match answer



Case 3: Invalid input

Action	Sent {"question":"hi"} to POST /api/assistant/questions using PowerShell.
Expected	HTTP 422 with the existing validation error.
Result	Pass. Status 422, INVALID_QUESTION, "Question must contain between 3 and 300 characters."

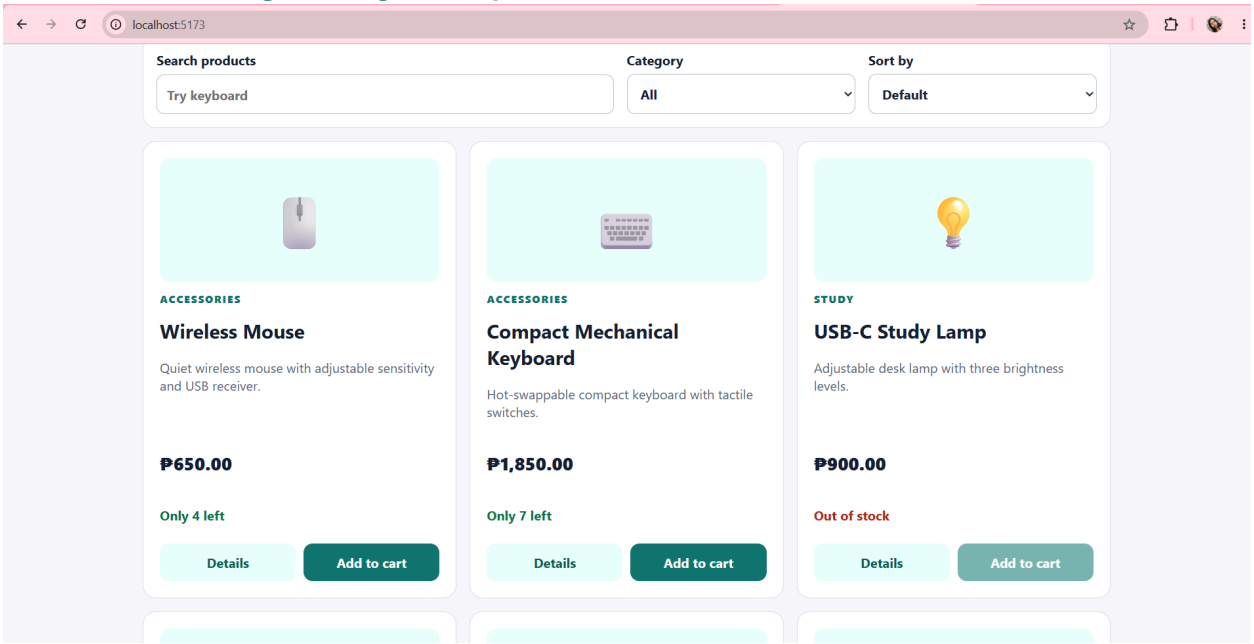
Screenshot: Powershell output



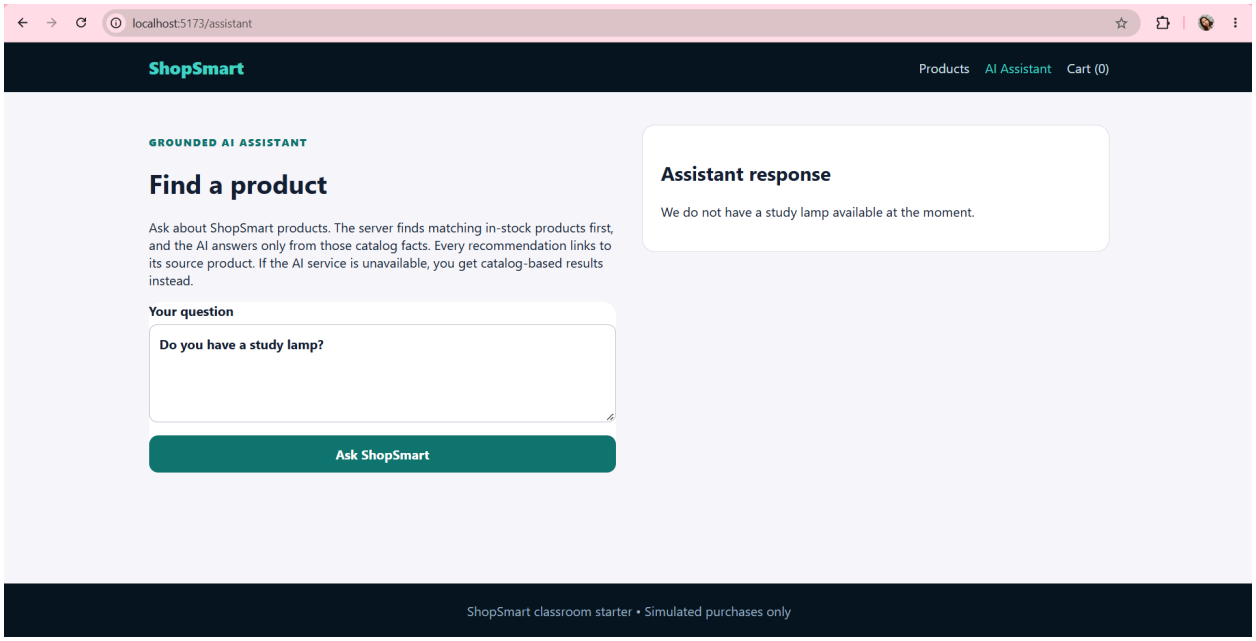
Case 4: Availability

Action	Asked "Do you have a study lamp?" The USB-C Study Lamp (id 3) has stock 0.
Expected	The stock-0 product is not recommended as available.
Result	Pass. "We do not have a study lamp available at the moment." The lamp was not shown.

Screenshot 1: catalog showing the lamp as "Out of stock"



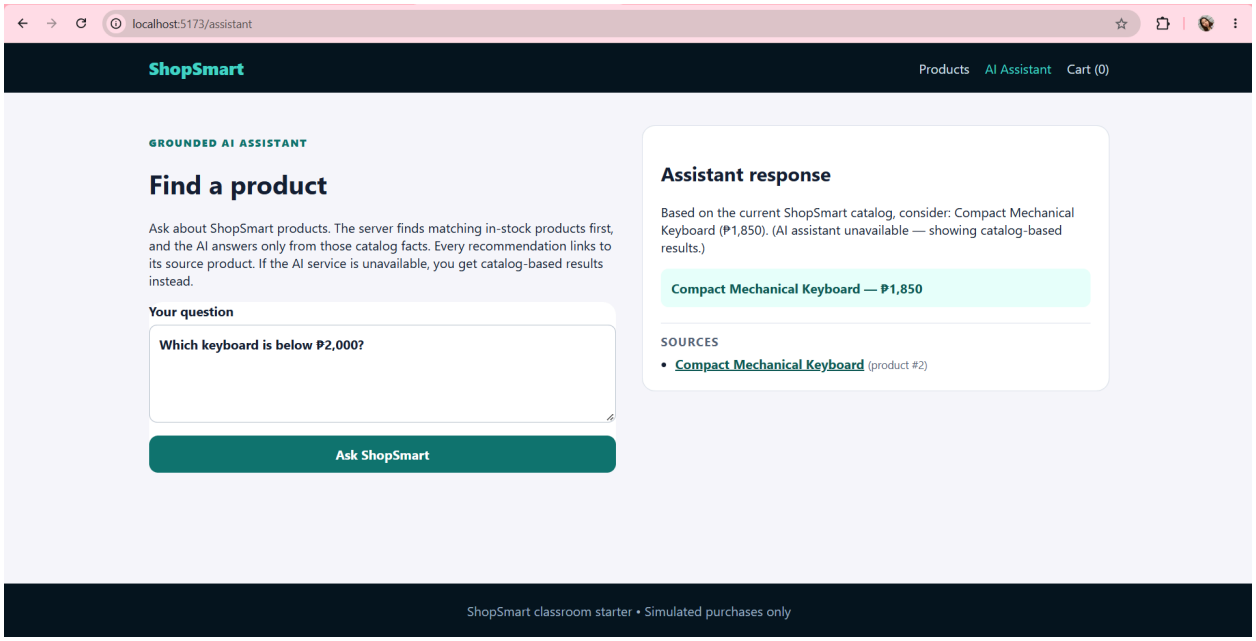
Screenshot 2: assistant answer



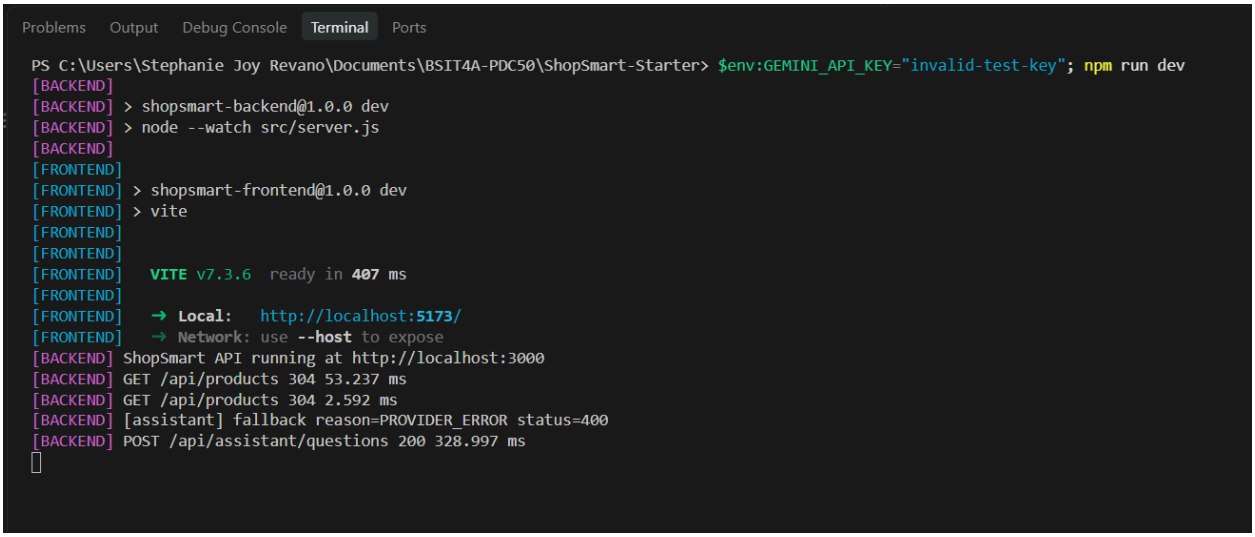
Case 5: Provider failure

Action	Started the app with an invalid key in the shell only: $env:GEMINI_API_KEY="invalid-test-key"; npm run dev. Then asked the keyboard question.
Expected	No endless loading. A clear message or rule-based fallback.
Result	Pass. Log showed "fallback reason=PROVIDER_ERROR status=400". Page showed the keyboard with "(AI assistant unavailable — showing catalog-based results.)" in about 329 ms.

Screenshot 1: assistant page with the fallback note



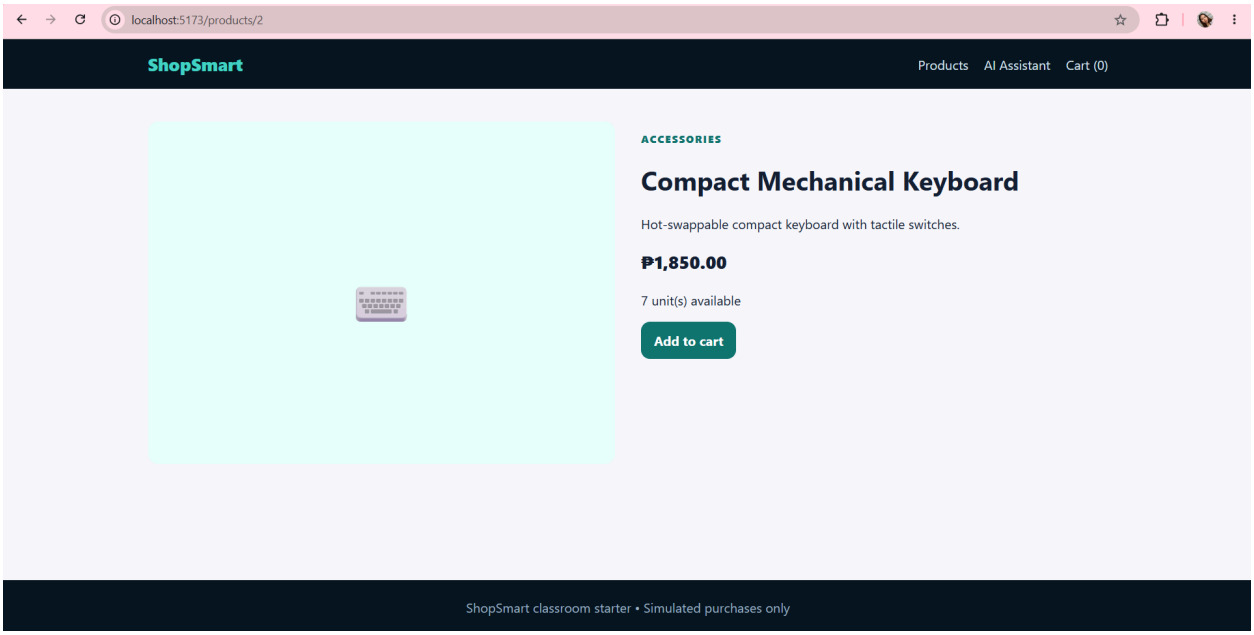
Screenshot 2: backend terminal log



Case 6: Source check

Action	Clicked the product link and the Sources link from Case 1.
Expected	Each link points to a product that was selected for this request.
Result	Pass. Both opened /products/2, which showed ₱1,850.00 and "7 unit(s) available", matching the answer.

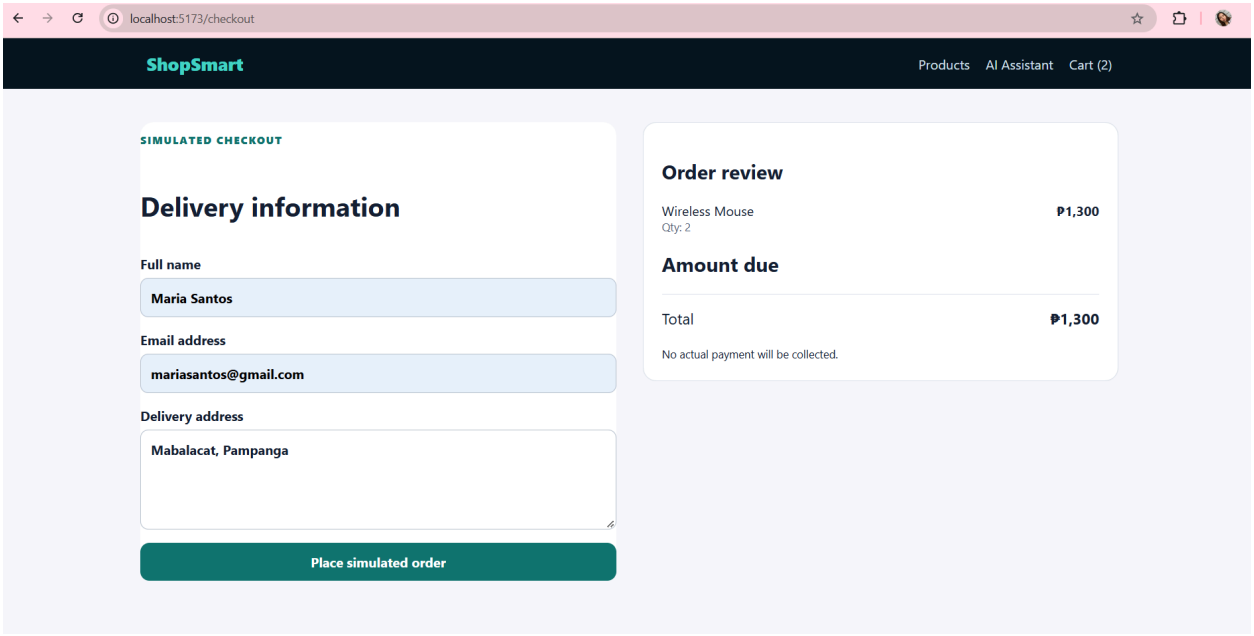
Screenshot: product page /products/2

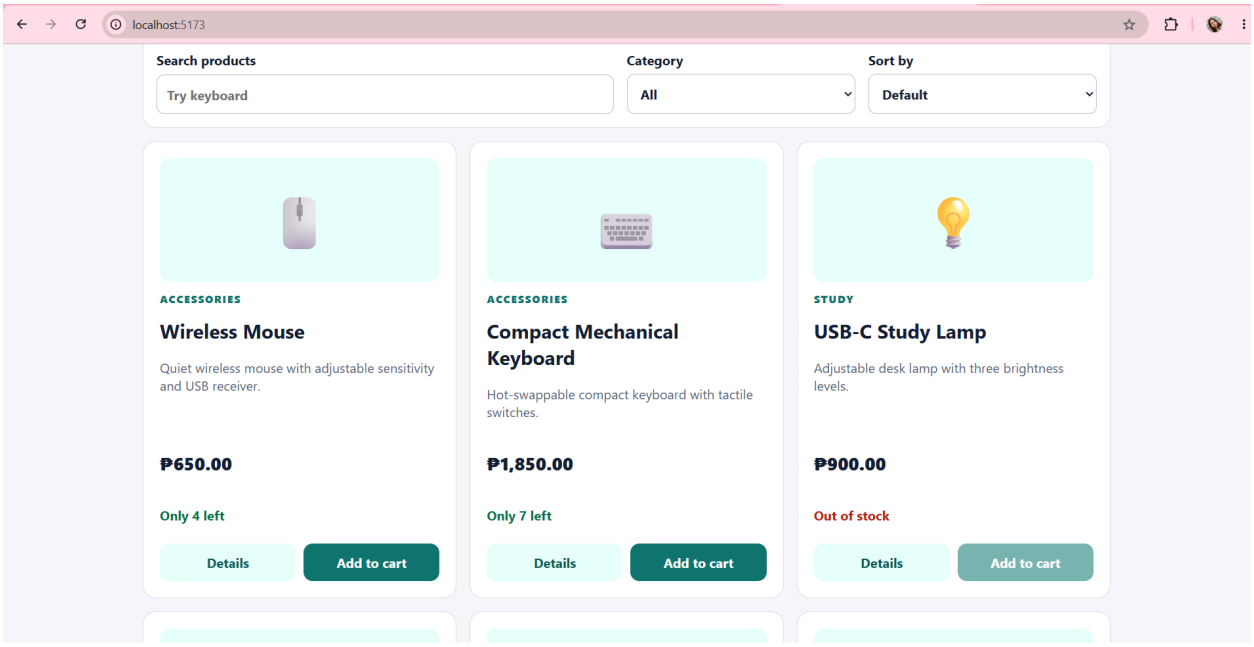


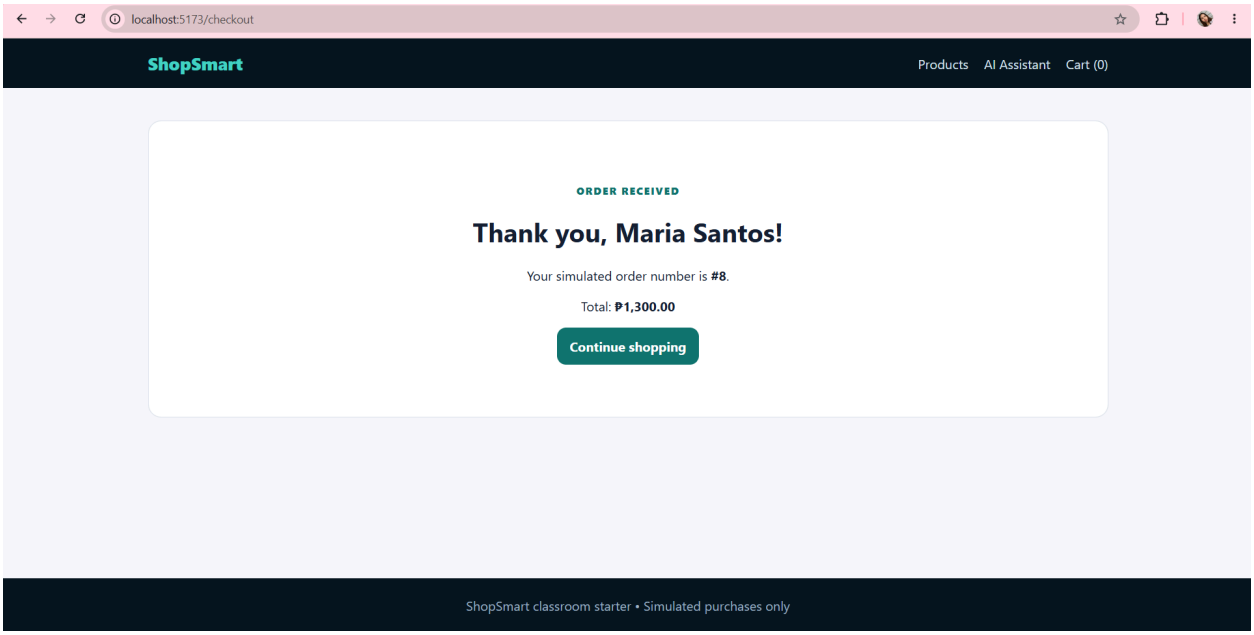
Checkout regression

Action	Added 2 Wireless Mouse to the cart and checked out.
Expected	Browsing, cart and checkout still work.
Result	Pass. The order went through and the Wireless Mouse stock went from 6 to 4.

Screenshot: products page showing "Only 4 left"






localhost:5173/checkout
ShopSmart
Products
AI Assistant
Cart (0)
ORDER RECEIVED
Thank you, Maria Santos!
Your simulated order number is #8.
Total: ₱1,300.00
Continue shopping
ShopSmart classroom starter • Simulated purchases only

npm run check

Action	Ran npm run check from the project root.
Expected	Lint, tests and build all pass.
Result	Pass. Lint passed. Frontend tests 11/11. Backend tests 58 pass, 0 fail. Build passed.

Screenshot: end of the npm run check output

